

# Komodo Dragon Element J

On May 1st, 2025, we had many experts from different fields come and evaluate our box. The Lead Herpetology Keeper, Cayton Curtis, and another Herpetology Keeper, James Flaherty, from the Nashville Zoo came to inspect our project. We have already communicated with these two when we visited the zoo earlier in the year to establish the problem. Along with them, the CCTE at Williamson County Schools, Jeremy Qualls, and Associate Professor of Engineering Education at Vanderbilt, Jeremi London, also came to help give us ideas. We explained the problem that we were trying to solve:

Nashville Zoo Keepers currently transport Komodo dragons in a black plastic tube that limits visibility and safety of the staff. Mr. Curtis and Mr. Flaherty already knew about the problem, but this was new information for Mr. Qualls and Dr. London.

After we explained the problem, we went on to show the different aspects of our boxes. We described the connection system that one of the other groups decided to create. This system uses butterfly clamps on the sides to connect the boxes and a PVC system to help support the three boxes when it is lifted up. Then we showed the handles and the frame and the medical door system that the last group has recently designed. All of our experts asked us great questions and gave some new feedback.

## EXPERT OPINIONS

Jeremi London gave us a great evaluation of our box. She did have a few suggestions.

|      |                                                |                                                            |                                                          |
|------|------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------|
| 3 /3 | Iteration of project design is not present.    | Partial iteration of project design is present             | Iteration of project design is present.                  |
| 3 /3 | Design is missing, unfinished, or plagiarized. | Design shows little evidence of creative/original thought. | Design shows some evidence of creative/original thought. |

Consider how much weight humans can safely carry.  
Add wheels for transport

Here shows her written feedback and her main two suggestions. For one thing, she suggested determining how much weight the humans can safely carry because if the box is heavy, then it will be too heavy with a large Komodo dragon in it. Another suggestion that we hadn't really thought of was adding wheels. This would make the weight issue not that much of a problem because one or two people could just roll the heavy box along the ground. The only issue with this is that the wheels would make it harder to keep the box in place while medical tests are being run on the Komodo dragon.

Jeremy Qualls also gave us some helpful feedback.

|      |                        |                        |                                     |
|------|------------------------|------------------------|-------------------------------------|
| 3 /3 | in the design process. | of the design process. | present and throughout the process. |
|------|------------------------|------------------------|-------------------------------------|

Total: 37 /42

Judge name: Qualls

Notes: Great job of working w/ the stakeholders on what was needed and iterations.

Total: 38 /42

Judge name: Qualls

Notes:

Great job w/ client (zoo) of listening to feedback and talking through iterations + flaws

Mr. Qualls was impressed with our ability to work with the client. As he expressed in both of his feedback sections, we worked well with the stakeholders as we had a zoom and in person meeting with them before the start of our product. He continues to detail that we explained our iterations and different parts of the boxes really well. He also noted that we were fine to talk about our flaws and shortcomings in the project.

#### ZOO EXPERT OPINIONS

The Lead Herpetology Keeper, Cayton Curtis, and another Herpetology Keeper at the zoo, James Flaherty, both gave our prototype a high rating (38/42 and 39/42 respectively).

Total: 38 /42

Judge name: Cayton Curtis

Notes:

**Total:** 39 /42

**Judge name:** James Plaherdy

**Notes:**

They both didn't write down any other feedback. He did tell us that he was a bit concerned with the flexibility of the plastic walls ( $\frac{1}{8}$ " ), but after we showed him how the plastic looked and swayed once we had bolted it in, he seemed much happier about the plastic.

Overall, we got great feedback from all of our experts and will use their information to continue to finish our prototype by May 8, when we deliver it to the zoo.